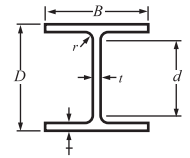


UNIVERSAL BEAMS AND COLUMNS



Section size	Unit weight M		Section area A		Section depth D		Flange width B		Flange thickness T		Web thickness t		Corner radius r	
in (mm)	lb/ft	kg/m	in ²	cm ²	in	mm	in	mm	in	mm	in	mm	in	mm
W16 16 x 7 (406 x 178)	36	54.0	10.6	68.39	15.86	402.8	6.985	177.4	0.43	10.90	0.295	7.49	0.40	10.2
	40	60.0	11.8	76.13	16.01	406.7	6.995	177.7	0.505	12.80	0.305	7.75	0.40	10.2
	45	67.0	13.3	85.81	16.13	409.7	7.035	178.7	0.565	14.30	0.345	8.76	0.40	10.2
	50	74.41	14.7	94.84	16.26	413.0	7.07	179.6	0.63	16.00	0.38	9.65	0.40	10.2
	57	85.0	16.8	108.4	16.43	417.3	7.12	180.8	0.715	18.16	0.43	10.92	0.40	10.2
W16 16 x 10 ¹ / ₄ (406 x 260)77	67	99.71	19.7	127.1	16.33	414.8	10.235	260.0	0.665	16.89	0.395	10.03	0.40	10.2
	114.6	22.6	145.8	16.52	419.6	10.295	261.5	0.76	19.30	0.455	11.56	0.40	10.2	10.2
	89	132.4	26.2	169.0	16.75	425.4	10.365	263.3	0.875	22.22	0.525	13.33	0.40	10.2
	100	148.8	29.4	189.7	16.97	431.0	10.425	264.8	0.985	25.02	0.585	14.86	0.40	10.2
W18 18 x 6 (457 x 152)	35	52.09	10.3	66.45	17.70	449.6	6.00	152.4	0.425	10.80	0.30	7.62	0.40	10.2
	40	60.00	11.8	76.13	17.90	454.7	6.015	152.8	0.525	13.30	0.315	8.00	0.40	10.2
	45	67.00	13.2	85.4	18.00	457.2	5.982	151.9	0.589	15.00	0.36	9.10	0.40	10.2
	46	68.46	13.5	87.1	18.06	458.7	6.06	153.9	0.605	15.30	0.36	9.14	0.40	10.2
	50	74.00	14.7	95.0	18.16	461.3	6.011	152.7	0.669	17.00	0.389	9.90	0.40	10.2
	55	82.00	16.2	104.5	18.31	465.1	6.042	153.5	0.744	18.90	0.42	10.70	0.40	10.2
W18 18 x 7 ¹ / ₂ (457 x 191)50	45	67.0	13.2	85.4	17.86	453.6	7.476	189.9	0.50	12.70	0.334	8.50	0.40	10.2
	74.41	14.7	94.84	17.99	456.9	7.495	190.4	0.57	14.40	0.355	9.02	0.40	10.2	10.2
	55	82.0	16.2	104.5	18.11	460.0	7.53	191.3	0.63	16.00	0.39	9.91	0.40	10.2
	60	89.29	17.6	113.5	18.24	463.3	7.555	191.9	0.695	17.65	0.415	10.54	0.40	10.2
	65	97.0	19.1	123.2	18.35	466.1	7.59	192.8	0.75	19.05	0.45	11.43	0.40	10.2
	66	98.0	19.4	125.3	18.40	467.4	7.592	192.8	0.77	19.60	0.45	11.40	0.40	10.2
	71	106.0	20.8	134.2	18.47	469.1	7.635	193.9	0.81	20.57	0.495	12.57	0.40	10.2
W18 18 x 11 (457 x 279)	76	113.1	22.3	143.9	18.21	462.5	11.035	280.3	0.68	17.27	0.425	10.80	0.40	10.2
	86	128.0	25.3	163.2	18.39	467.1	11.09	281.7	0.77	19.56	0.48	12.19	0.40	10.2
	97	144.4	28.5	183.9	18.59	472.2	11.145	283.1	0.87	22.10	0.535	13.59	0.40	10.2
	106	157.7	31.1	200.6	18.73	475.7	11.20	284.5	0.94	23.88	0.59	14.99	0.40	10.2
	119	177.1	35.1	226.5	18.97	481.8	11.265	286.1	1.06	26.92	0.655	16.64	0.40	10.2
W21 21 x 6 ¹ / ₂ (533 x 165)50	44	66.0	13.0	83.87	20.66	524.8	6.50	165.1	0.45	11.40	0.35	8.89	0.50	12.7
	74.41	14.7	94.84	20.83	529.1	6.53	165.9	0.535	13.50	0.38	9.65	0.50	12.7	12.7
	57	85.0	16.7	107.7	21.06	534.9	6.555	166.5	0.65	16.51	0.405	10.29	0.50	12.7
W21 21 x 8 ¹ / ₄ (533 x 210)62	55	82.0	16.2	104.4	20.80	528.3	8.216	208.7	0.52	13.20	0.376	9.60	0.50	12.7
	92.27	18.3	118.1	20.99	533.1	8.24	209.3	0.615	15.62	0.40	10.16	0.50	12.7	12.7
	68	101.2	20.0	129.0	21.13	536.7	8.27	210.1	0.685	17.40	0.43	10.92	0.50	12.7
	73	109.0	21.5	138.7	21.24	539.5	8.295	210.7	0.74	18.80	0.455	11.56	0.50	12.7
	82	122.0	24.1	155.8	21.44	544.6	8.342	211.9	0.84	21.30	0.502	12.80	0.50	12.7
	83	123.5	24.3	156.8	21.43	544.3	8.355	212.2	0.835	21.21	0.515	13.08	0.50	12.7
	93	138.4	27.3	176.1	21.62	549.1	8.42	213.9	0.93	23.62	0.58	14.73	0.50	12.7
W21 21 x 12 ¹ / ₄ (533 x 311)111	101	150.3	29.8	192.3	21.36	542.5	12.29	312.2	0.80	20.32	0.50	12.70	0.50	12.7
	165.2	32.7	211.0	21.51	546.4	12.34	313.4	0.875	22.22	0.55	13.97	0.50	12.7	12.7
	122	181.6	35.9	231.6	21.68	550.7	12.39	314.7	0.96	24.38	0.60	15.24	0.50	12.7
	132	196.4	38.8	250.3	21.83	554.5	12.44	316.0	1.03	26.29	0.65	16.51	0.50	12.7
	147	218.8	43.2	278.7	22.06	560.3	12.51	317.8	1.15	29.21	0.72	18.29	0.50	12.7
W24 24 x 7 (610 x 178)	55	82	16.2	104.5	23.57	598.7	7.005	177.9	0.505	12.83	0.395	10.03	0.50	12.7
	61	91	17.9	116.5	23.72	602.5	7.023	178.4	0.591	15.00	0.419	10.60	0.50	12.7
	62	92.27	18.2	117.4	23.74	603.0	7.040	178.8	0.59	14.99	0.43	10.92	0.50	12.7